Free-body diagrams, part 1

LEARNING TARGET · S5 I can draw a complete FBD — every force, with arrows sized to strength.

A · KEY VOCABULARY

Free-body diagram	Only the forces acting *on* one chosen object.
The dot / box	Replace the object with a simple shape; draw arrows from it.
Source	Every arrow needs one: rope, floor, table, Earth.
No "motion force"	Motion is not an interaction. Leave velocity off the diagram.

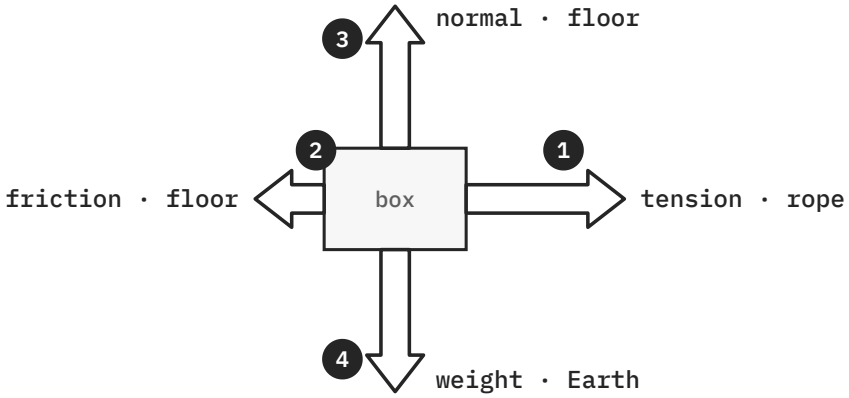
B · SIX FBD RULES

1 just the object (a dot)
2 one arrow per force
3 arrow points the force's way
4 length ∝ strength
5 label each: F_g, F_N, F_f…
6 only forces ON the object

routine choose · list · draw

C · COLOR IT · A BOX PULLED RIGHT

Rough level floor, rope pulls right.



- ☐ **1 · ORANGE** Tension (T)
- ☐ **2 · ORANGE** Friction (f)
- ☐ **3 · ORANGE** Normal force (N)
- ☐ **4 · ORANGE** Weight (W)

D · WORKED EXAMPLE

A 20 N lamp hangs at rest from one vertical rope. Draw and label its forces.

GIVEN	EQUATION	WORK	ANSWER
weight = 20 N at rest one rope	sources: Earth, rope	weight 20 N down tension up, balances it	**Tension 20 N up** equal-length arrows

E · THE 2026-XJ THREAD

DAY 15 OF 22 · NEW TOOL: FREE-BODY DIAGRAMS

A free-body diagram puts XJ on a dot and shows every force acting on it. For an object in space the list is short, which is what makes it predictable.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else